Lazizbek Sadullaev. Math 464. HW #4.

For the given problem, we need to optimize the number of trees to be planted that satisfy the constraints given in the problem.

Let x_1 be the number of apples
x_2 be the number of pears
x_3 be the number of cherries.

So the optimization problem is as follows:

$$\max \quad z = x_1 + x_2 + x_3 + 0 \cdot y_1 + 0 \cdot y_2 + 0 \cdot y_3$$

s.t.

$$x_1 + x_2 + x_3 \leq 345 \rightarrow \text{max. \# of trees}$$

$$150\, x_1 + 200\, x_2 + 240\, x_3 \leq 50.000 \rightarrow \text{Labor cost constrain}$$

$$275\, x_1 + 180\, x_2 + 125\, x_3 \leq 60.000 \rightarrow \text{material cost constrain}$$

(1)

$$150\, y_1 \leq x_1 \leq 345\, y_1 \quad \text{where } y_1 \in \{0, 1\}$$

$$50 \cdot y_2 \leq x_2 \leq 345\, y_2 \quad \text{where } y_2 \in \{0, 1\}$$

$$80\, y_3 \leq x_3 \leq 345\, y_3 \quad \text{where } y_3 \in \{0, 1\}$$

$$x = [x_1, x_2, x_3]^T \geq 0$$

$$x \in \mathbb{Z}^3.$$

Now, let me try to adjust this maximization problem so that program (software) can understand:

We have new (additional) decision variables y_1, y_2, y_3 to satisfy the problem requirements. But these variables (y_1, y_2, y_3) are not participating in the objective function $z = x_1 + x_2 + x_3$, so I added them with <u>zero</u> coefficients to make them appear in z. Now, (i) we should do the same thing to constrain inequalities, and (ii) write the last inequalities in the standard format. Let's do these 2 steps:

(i) $150x_1 + 200x_2 + 240x_3 \leq 50.000$ → Labor constrain

$\Downarrow$

(2) $150x_1 + 200x_2 + 240x_3 + 0 \cdot y_1 + 0 \cdot y_2 + 0 \cdot y_3 \leq 50.000$

similarly

$275x_1 + 180x_2 + 125x_3 \leq 60.000$ → material constrain

$\Downarrow$

(3) $275x_1 + 180x_2 + 125x_3 + 0 \cdot y_1 + 0 \cdot y_2 + 0 \cdot y_3 \leq 60.000$

(ii) For the last 3 constrains in (1) where we used binary desicion variables, let's write them in the standard form similar to (2) & (3):

$$150 y_1 \le x_1 \le 345 y_1 \quad \begin{cases} 150 y_1 \le x_1 & (4) \\ x_1 \le 345 y_1 & (5) \end{cases}$$

(4) can be written as:

$150 y_1 - x_1 \le 0$ & making it standard

(6) $-x_1 + 0 \cdot x_2 + 0 \cdot x_3 + 150 y_1 + 0 \cdot y_2 + 0 \cdot y_3 \le 0$

Similarly, (5) can be written as:

$x_1 - 345 y_1 \le 0$ & making is standard:

(7) $x_1 + 0 x_2 + 0 \cdot x_3 - 345 y_1 + 0 \cdot y_2 + 0 \cdot y_3 \le 0$

So the following substitution is valid:

$$150 y_1 \le x_1 \le 345 y_1 \iff \begin{cases} (6)\ -x_1 + 0 \cdot x_2 + 0 \cdot x_3 + 150 y_1 + 0 \cdot y_2 + 0 \cdot y_3 \le 0 \\ \& \\ (7)\ x_1 + 0 \cdot x_2 + 0 \cdot x_3 - 345 y_1 + 0 \cdot y_2 + 0 \cdot y_3 \le 0 \end{cases}$$

(8) where $0 \le y_1 \le 1$ & $y_1 \in \mathbb{Z}$

We can write the remaing 2 constraints in a similar fashion.

$$50 y_2 \leq x_2 \leq 345 y_2 \qquad y_2 \in \{0,1\} \iff$$

$$\iff \quad 0 \cdot x_1 - x_2 + 0 \cdot x_3 + 0 \cdot y_1 + 50 y_2 + 0 \cdot y_3 \leq 0$$

$$(9) \quad 0 \cdot x_1 + x_2 + 0 \cdot x_3 + 0 \cdot y_1 - 345 y_2 + 0 \cdot y_3 \leq 0$$

where $0 \leq y_2 \leq 1 \qquad y_2 \in \mathbb{Z}$,

and finally

$$80 y_3 \leq x_3 \leq 345 y_3 \quad , \quad y_3 \in \{0,1\} \iff$$

$$0 \cdot x_1 + 0 \cdot x_2 - x_3 + 0 \cdot y_1 + 0 \cdot y_2 + 80 y_3 \leq 0$$

$$\iff$$

$$(10) \quad 0 \cdot x_1 + 0 \cdot x_2 + x_3 + 0 \cdot y_1 + 0 \cdot y_2 - 345 y_3 \leq 0$$

where $0 \leq y_3 \leq 1 \quad , \quad y_3 \in \mathbb{Z}$.

By replacing these 3 constraints by their respective equivalent constraints the maximal optimization problem (1) becomes

$$\max_{x,y} \quad z = x_1 + x_2 + x_3 + 0 \cdot y_1 + 0 \cdot y_2 + 0 \cdot y_3$$

s.t.

$$
\begin{aligned}
x_1 + x_2 + x_3 + 0 \cdot y_1 + 0 \cdot y_2 + 0 \cdot y_3 &\leq 345 \\
150 x_1 + 200 x_2 + 240 x_3 + 0 \cdot y_1 + 0 \cdot y_2 + 0 \cdot y_3 &\leq 50.000 \\
275 x_1 + 180 x_2 + 125 x_3 + 0 \cdot y_1 + 0 \cdot y_2 + 0 \cdot y_3 &\leq 60.000 \\
-x_1 + 0 \cdot x_2 + 0 \cdot x_3 + 150 y_1 + 0 \cdot y_2 + 0 \cdot y_3 &\leq 0 \\
x_1 + 0 \cdot x_2 + 0 \cdot x_3 - 345 y_1 + 0 \cdot y_2 + 0 \cdot y_3 &\leq 0 \\
0 \cdot x_1 - x_2 + 0 \cdot x_3 + 0 \cdot y_1 + 50 y_2 + 0 \cdot y_3 &\leq 0 \\
0 \cdot x_1 + x_2 + 0 \cdot x_3 + 0 \cdot y_1 - 345 y_2 + 0 \cdot y_3 &\leq 0 \\
0 \cdot x_1 + 0 \cdot x_2 - x_3 + 0 \cdot y_1 + 0 \cdot y_2 + 80 y_3 &\leq 0 \\
0 \cdot x_1 + 0 \cdot x_2 + x_3 + 0 \cdot y_1 + 0 \cdot y_2 - 345 y_3 &\leq 0
\end{aligned}
\tag{11}
$$

$$x \geq 0$$

$$0 \leq y \leq 1$$

$$x \in \mathbb{Z}^3, \qquad y \in \mathbb{Z}^3$$

I'll be solving this (11) optim. problem by software and attaching the results separately:

$$x^* = [\overset{x_1}{172}, \overset{x_2}{0}, \overset{x_3}{100}, \overset{y_1}{1}, \overset{y_2}{0}, \overset{y_3}{1}]^T \quad \&$$

$$z^* = 172 + 100 = 272.$$